

David Yifan Yao

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Education

University of Illinois Urbana-Champaign (UIUC)

M.S. in Computer Science (GPA: 4.0/4.0)

Aug. 2023 - May. 2025

Advisor: Shenlong Wang

University of Illinois Urbana-Champaign (UIUC)

B.S. in Mathematics and Computer Science with minor in Statistics (GPA: 4.0/4.0)

Aug. 2020 - May. 2023

Publications (* indicates equal contribution and † indicates equal advising)

[1] Uni4D: Unifying Large Vision Models for 4D Modeling from a Single Video

David Yifan Yao, Albert J. Zhai, Shenlong Wang
CVPR (Highlight), 2025

[2] Visual Sync: Multi-Camera Synchronization via Cross-View Object Motion

Shaowei Liu*, David Yifan Yao*, Saurabh Gupta†, Shenlong Wang†
NeurIPS, 2025

Research Experience

Graduate Research Assistant at Shenlong's Lab, UIUC

May. 2023 - May. 2025

Advisor: Prof. Shenlong Wang

- Reconstruct accurate pose, geometry, and motion from a single casual monocular video. [1]
- Developed an optimization framework based on multi-view dynamics that aligns unposed, unsynchronized in-the-wild videos at millisecond accuracy [2]
- Experiment with lifting foundational 2D image features into 3D for point cloud instance segmentation

Research Assistant at Forward Data Lab, UIUC

Jun. 2022 - Feb. 2023

Advisor: Prof. Kevin Chang

- Analyzed characteristics and applicability of various word embedding models such as BERT, CBOW, Skip-gram, and Flair in clustering

Work Experience

Tesla Autopilot

May 2025 - Present

Machine Learning Engineer on Geometric Vision Team

Palo Alto, US

- Experimented and shipped autolabel with novel techniques, improving labeling efficiency by **50%**

Software Engineer intern on Audio Visual Team

May 2022 - Aug 2022

Roblox

San Mateo, US

- Kickstarted first research project on automated solutions for video moderation
- Performed extensive literature review and proof-of-concept experimentation with video classification and Out-Of-Distribution detection algorithms
- Independently proposed, implemented, and deployed real-time violence detector into video upload pipeline with **77%** AUC

Data Science Intern

May 2021 - Aug 2021

SmartClean

Singapore

- Deployed pre-trained object detection model with TensorFlow Lite on Raspberry Pi to detect litter using IP cameras
- Experimented with various audio signal processing methods on raw audio for audio classification

Data Science Intern

Feb 2020 - Jun 2020

Ai Palette

Singapore

- Data collection through web scraping online forums and managing with Elasticsearch
- Text training data tokenization and preprocessing for semantic analysis
- Time series forecasting using basic statistical techniques.

Honors & Awards

2024	Bronze Tablet, Class of 2025 , - top 3% of graduating cohort	UIUC
2020-2023	Dean's List , - Top 20% GPA every semester	UIUC

Teaching Experience (UIUC)

CS445 Computational Photography *Spring 2025*
Instructor: Prof. Derek Hoiem

CS441 Applied Machine Learning *Fall 2024*
Instructor: Prof. Marco Morales Aguirre

CS543 Computer Vision *Spring 2024*
Instructor: Prof. Saurabh Gupta

CS441 Applied Machine Learning *Fall 2023*
Instructor: Prof. Marco Morales Aguirre

CS473 Algorithms *Spring 2023*
Instructor: Prof. Timothy Chan

Extracurricular Activity

2021-2022	Member , UIUC Men's Ultimate Frisbee Team (D1)	UIUC
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